Size Transferability of Graph Transformers with Convolutional Positional Encodings

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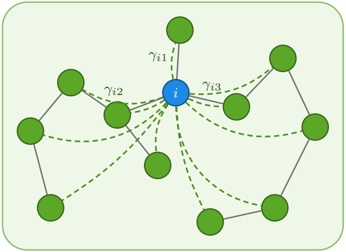
Why size-transferability?

- Graph data is scarce and computationally expensive to process $\rightarrow$ Large scale Graph ML requires *efficient* and *scalable* training.
- **Size transferability:** ability of an architecture to generalize to larger graphs than those observed during training.
- GNNs are provably size-transferable [Ruiz et al., 2020, 2023, Keriven et al., 2020, Wang et al., 2024, Maskey et al., 2023]. What about Graph Transformers (GT)?
- **Idea:** Can we leverage a transferable architecture to learn transferable positional encodings for GT?

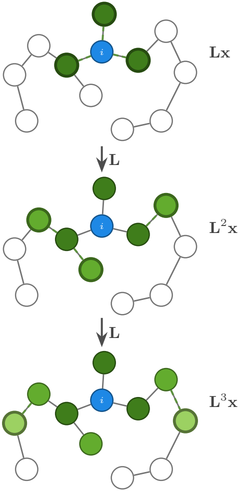
Claim: GTs *inherit* size-transferability from their positional encodings.

Attention vs Graph Convolution



$$\mathbf{y}_i = \frac{\sum_{j=1}^{N} \gamma_{ij}\, \mathbf{V}\mathbf{x}_j}{\sum_{j=1}^{N} \gamma_{ij}}$$

Attention



$$\mathbf{y} = \sum_{k=0}^{K} h_k\, \mathbf{L}^k \mathbf{x}$$

Graph Convolution

Preliminaries: manifolds and graph construction

- **Manifold model:** $\mathcal{M} \subset \mathbb{R}^{\mathrm{M}}$ is a d-dim. compact, smooth submanifold with induced measure μ.
- **Geometric graph construction:** Sample manifold points $X_N = \{x_i\}_{i=1}^{N}$ according to distribution μ. Construct graph $\mathbf{G}_N$ by connecting (x_i, x_j) with edge weights $\mathcal{W}_{ij} = K_\epsilon \frac{\|x_i - x_j\|}{\epsilon}$.
- **Signals:** manifold signal $f \in L^2(\mu; \mathbb{R}^D)$, sampling yields graph signals $\mathbf{S}_N f = \mathbf{X} \in \mathbb{R}^{N \times D}$.
- Graph Laplacian converges to the Laplace-Beltrami operator, $\mathbf{L}_N \to \mathcal{L}$ [Belkin and Niyogi, 2008].

Preliminaries: Graph Transformer with Positional Encodings

Definition — Generalized Graph Transformer

A GT layer is given by

$$\Phi_{\mathbf{G}}(\mathcal{T}, \mathbf{X})_i = \mathbf{y}_i = \frac{\sum_{j=1}^{N} \gamma_{ij} \, \mathbf{V}\mathbf{x}_j}{\sum_{j=1}^{N} \gamma_{ij}}.$$

- **Kernel function:** $\gamma : L^2(\mu; \mathbb{R}^D) \times L^2(\mu; \mathbb{R}^D) \to \mathbb{R}$.
  - **Example:** Softmax attention $\gamma_{ij} = \exp \langle \mathbf{Q}\mathbf{x}_i, \mathbf{K}\mathbf{x}_j \rangle$
- **Input:** $\mathbf{X} = \mathbf{X}_{\text{in}} + \boldsymbol{\Psi}_{\mathbf{G}}(\mathcal{H}, \mathbf{L}, \mathbf{Z})$, where $\boldsymbol{\Psi}_{\mathbf{G}} : \mathbb{R}^{N \times D} \to \mathbb{R}^{N \times D}$ is the *positional encoder*.
  - **Examples:** GNNs [Kanatsoulis et al., 2024], Laplacian eigenvectors [Dwivedi and Bresson, 2020], Random Walk PEs [Dwivedi et al., 2021]

Definition: Manifold Transformer

Definition — Manifold Transformer

A manifold transformer (MT) layer is given by, for all $x \in \mathcal{M}$,

$$\Phi_{\mathcal{M}}(\mathcal{T}, f)(x) = \frac{\int_{\mathcal{M}} \tilde{\gamma}_{xy}\, \mathbf{V} f(y)\, d\mu(y)}{\int_{\mathcal{M}} \tilde{\gamma}_{xy}\, d\mu(y)}.$$

- **Kernel:** $\tilde{\gamma}_{xy} := \gamma(f(x), f(y))$ continuous analogue of the GT kernel γ_{ij}.
- **Input:** $f(x) = \Psi_{\mathcal{M}}(\mathcal{H}, \mathcal{L}, g)(x)$, the limit positional encoding, with input manifold signal $g \in L^2(\mathcal{M})$.
- Attention integrated over the manifold characterizes the limit of the generalized GT.

Definition: Transferability

Definition: Transferability

$\boldsymbol{\Psi}$ is *size-transferable* if, for every $f \in L^2(\mu; \mathbb{R}^p)$,

$$\lim_{N\to\infty} \left\| \mathbf{I}_N \boldsymbol{\Psi}_{\mathbf{G}}(\mathbf{L}_N, \mathbf{S}_N f) - \boldsymbol{\Psi}_{\mathcal{M}}(\mathcal{L}, f) \right\|_{L^2(\mu)} = 0.$$

- **Sampling operator $\mathbf{S}_N$** $: L^2(\mu; \mathbb{R}^p) \to \mathbb{R}^{N\times p}$ — evaluates a manifold signal on the N sampled nodes.
- **Interpolation operator $\mathbf{I}_N$** $: \mathbb{R}^{N\times p} \to L^2(\mu; \mathbb{R}^p)$ — lifts a graph signal back to the manifold.
- Bounded and consistent: $\lim_{N\to\infty} \|\mathbf{I}_N \mathbf{S}_N f - f\|_{L^2(\mu)} = 0$.

GT with transferable PEs

Graph Transformer

$$\underbrace{[\boldsymbol{\Psi}_{\mathbf{G}}(\mathcal{H}, \mathbf{L}, \mathbf{Z})]_i}_{\text{graph PE}} = \mathbf{x}_i$$

$$[\boldsymbol{\Phi}_{\mathbf{G}}(\mathcal{T}, \mathbf{X})]_i = \frac{\sum_{j=1}^{N} \gamma_{ij} \mathbf{V}\mathbf{x}_j}{\sum_{j=1}^{N} \gamma_{ij}}$$

$$\xrightarrow{N \to \infty}$$

Manifold Transformer

$$\underbrace{\boldsymbol{\Psi}_{\mathcal{M}}(\mathcal{H}, \mathcal{L}, g)(x_i)}_{\text{manifold PE}} = f(x_i)$$

$$\boldsymbol{\Phi}_{\mathcal{M}}(\mathcal{T}, f)(x_i) = \frac{\int_{\mathcal{M}} \tilde{\gamma}_{x_i y}\, \mathbf{V} f(y)\, d\mu(y)}{\int_{\mathcal{M}} \tilde{\gamma}_{x_i y}\, d\mu(y)}$$

Manifold Transformer (MT) is the limit object for GT as graph size increases.

Softmax GT with GNN PEs

Graph Transformer

$$[\mathbf{\Psi}_{\mathbf{G}}(\mathcal{H}, \mathbf{L}, \mathbf{Z})]_i = \mathbf{x}_i = \left[\sigma\left(\sum_{k=0}^{K-1} \mathbf{H}_k \mathbf{Z} \mathbf{L}^k\right)\right]_i$$

$$[\mathbf{\Phi}_{\mathbf{G}}(\mathcal{T}, \mathbf{X})]_i = \frac{\sum_{j=1}^{N} e^{\langle \mathbf{Q}\mathbf{x}_i, \mathbf{K}\mathbf{x}_j \rangle} \mathbf{V}\mathbf{x}_j}{\sum_{j=1}^{N} e^{\langle \mathbf{Q}\mathbf{x}_i, \mathbf{K}\mathbf{x}_j \rangle}}$$

$$\xrightarrow{N \to \infty}$$

Manifold Transformer

$$\mathbf{\Psi}_{\mathcal{M}}(\mathcal{H}, \mathcal{L}, g)(x_i) = f(x_i) = \sigma\left(\sum_{k=0}^{K-1} h_k \, e^{-k\mathcal{L}} g(x_i)\right)$$

$$\mathbf{\Phi}_{\mathcal{M}}(\mathcal{T}, f)(x_i) = \frac{\int_{\mathcal{M}} e^{\langle \mathbf{Q} f(x_i), \mathbf{K} f(y) \rangle} \mathbf{V} f(y) \, d\mu(y)}{\int_{\mathcal{M}} e^{\langle \mathbf{Q} f(x_i), \mathbf{K} f(y) \rangle} \, d\mu(y)}$$

- **Concrete example:** GT with GNN PEs converges to MT with MNN PEs

GNNs are nonlinear functions of eigenvectors

Proposition 3.1 [Kanatsoulis et al., 2024]

A GNN (...) operates as a nonlinear function of the GSO eigenvectors i.e., $\boldsymbol{x}_v^{(l)} = \mathtt{MLP}\left(\boldsymbol{v}^{(v)}\right)$, $\boldsymbol{v}^{(v)} = \boldsymbol{V}[v,:]^T$. The trainable parameters of the first MLP layer (...) depend on the eigenvalues $\{\lambda_n\}_{n=1}^N$ and eigenvectors $\{\boldsymbol{v}_n\}_{n=1}^N$ of the GSO, as well as the node features $\boldsymbol{X}$ of the graph:

$$\boldsymbol{x}_v^{(l)} = \mathtt{MLP}\left(\boldsymbol{v}^{(v)}\right) = \mathtt{MLP}^{(-1)}\left(\sigma\left(\boldsymbol{W}^T\boldsymbol{v}^{(v)}\right)\right) \tag{1}$$

$$\boldsymbol{W}[n,f] = \sum_{i=1}^{F_{l-1}}\sum_{k=0}^{K-1} \lambda_n^k \boldsymbol{H}_k^{(l)}[i,f]\langle \mathbf{v}_n, \boldsymbol{X}^{(l-1)}[:,i]\rangle, \tag{2}$$

Architecture: Graph Transformer with RPEARL PE

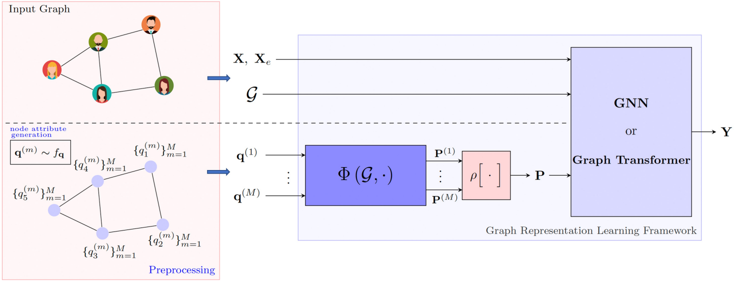


Figure: RPEARL Architecture Diagram[Kanatsoulis et al., 2024]

- Improved GNN expressive power in the WL sense.
- Recover permutation equivariance via pooling over M realizations.

Is GT with GNN-based (RPEARL) positional encodings size transferable?

Standard assumptions

Assumption — Normalized Lipschitz Signals

The input manifold signals g are normalized Lipschitz: for all $a, b \in \mathcal{M}$, $\|g(b) - g(a)\| \leq \|b - a\|$.

Assumption — Bounded Linear Operators

$\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are bounded linear operators with constants $C_Q, C_K, C_V > 0$:

$$\|\mathbf{Qx}\| \leq C_Q\|\mathbf{x}\|, \quad \|\mathbf{Kx}\| \leq C_K\|\mathbf{x}\|, \quad \|\mathbf{Vx}\| \leq C_V\|\mathbf{x}\|.$$

- Mild assumption on the manifold signals and transformer parameters.

Assumption: Valid kernel function

Assumption — Valid Kernel Function

The kernel function $\gamma_{xy} = \gamma(f(x), f(y))$ satisfies:

1. **Nonnegative:** $\gamma_{xy} \geq 0$ for any $x, y \in \mathcal{M}$.
2. **Bounded:** $\gamma_{xy} \leq C$ for some constant C.
3. For $x, y, x', y' \in \mathcal{M}$ and signal f,

$$\|\gamma_{xy} - \gamma_{x'y'}\| \leq L_\gamma(\|f(x) - f(x')\| + \|f(y) - f(y')\|).$$

- Requires that the kernel function is well-defined.
- Readily satisfied for the softmax kernel.

Assumption: Transferable PE

Assumption — Transferable Positional Encoding

The PE $\mathbf{\Psi}_{\mathbf{G}}(\mathcal{H}, \mathbf{X})$ converges to a limit object $\mathbf{\Psi}_{\mathcal{M}}(\mathcal{H}, f) : \mathcal{M} \to \mathbb{R}^q$, with

$$\|\mathbf{\Psi}_{\mathbf{G}}(\cdot)(x) - \mathbf{\Psi}_{\mathcal{M}}(\cdot)(x)\| \leq \Delta_{\mathrm{PE}}.$$

- Satisfied when $\mathbf{\Psi}_{\mathbf{G}}$ is a GNN [Wang et al., 2024].
- Can also be shown for other graph Laplacian-based encodings, e.g., Laplacian eigenvectors, random walk PEs, etc.

Theorem: GT to MT output convergence

Theorem — Point-wise Convergence of GT to MT

Under Assumptions 1–4, for any $x \in \mathcal{M}$, w.p. $1-\delta$,

$$\begin{aligned}\|\boldsymbol{\Phi}_{\mathbf{G}}(\mathcal{T},\mathbf{X})(x) - \boldsymbol{\Phi}_{\mathcal{M}}(\mathcal{T},f)(x)\|_2 &\leq \Delta_{\boldsymbol{\Phi}(\mathbf{X})} \\ &= \left[C_V + 4e^{2M} C_{\mathrm{QKV}}\right] \Delta_{\mathrm{PE}} \\ &\quad + \left[e^{M} C_V + 2e^{2M} C_{\mathrm{QKV}}\right] A\left(\frac{\log N}{N}\right)^{1/d}\end{aligned}$$

- Output difference of GT to MT vanishes with rate depending on
  - **Positional-encoding error** Δ_{PE}: output difference between graph PE and manifold PE.
  - **Operator bound constants**: smoother filters improve convergence.
  - **Kernel bound**: for softmax, $M = \sup_{u,v} \langle \mathbf{Q}f(u), \mathbf{K}f(v)\rangle$.
  - Order $\mathcal{O}\left(\left(\frac{\log N}{N}\right)^{1/d}\right)$ and A dependent on manifold geometry.

Corollary: GT transferability

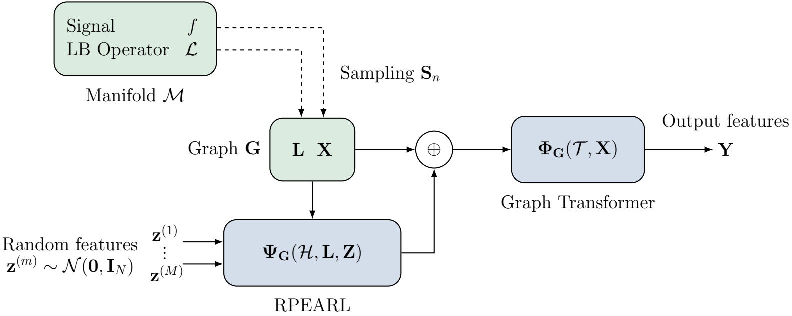
Corollary 1 — Transferability of GTs

For graphs $\mathbf{G}_1, \mathbf{G}_2$ sampled from $\mathcal{M}$ with N_1, N_2 nodes and signals $\mathbf{X}_1, \mathbf{X}_2$, w.p. $1-\delta$,

$$\frac{1}{\mu(\mathcal{M})} \left\| \mathbf{I}_{N_1} \mathbf{\Phi}_{\mathbf{G}_1}(\,\cdot\,) - \mathbf{I}_{N_2} \mathbf{\Phi}_{\mathbf{G}_2}(\,\cdot\,) \right\|_{L^{1,2}(\mathcal{M})} \leq \Delta_{\mathbf{\Phi}(\mathbf{X}_1)} + \Delta_{\mathbf{\Phi}(\mathbf{X}_2)} + 4\, e^{2M}\, C_{\mathrm{QKV}}\, A \left(\frac{\log N}{N} \right)^{1/d}.$$

- $\Delta_{\mathbf{\Phi}(\mathbf{X}_1)}$, $\Delta_{\mathbf{\Phi}(\mathbf{X}_2)}$: each graph's GT-to-MT output difference (Theorem 1).
- Sampling term decays with graph sizes.
- **Takeaway:** Train a GT on a small graph $\mathbf{G}_1$, transfer to a larger $\mathbf{G}_2$ ($N_2 > N_1$) with bounded output difference.

Transferability of GT with RPEARL positional encodings.



$$\mathbf{\Psi}_{\mathbf{G}}(\mathcal{H}, \mathbf{L}, \mathbf{Z}) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{P}^{(m)}, \qquad \mathbf{P}^{(m)} = \sigma\left[\sum_{k=0}^{K-1} \mathbf{H}_k \mathbf{z}^{(m)\top} \mathbf{L}^k \right],$$

Corollary: RPEARL Transferability

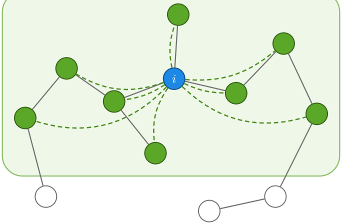
Corollary 2 — Transferability of GT with RPEARL PEs

For the softmax GT $\mathbf{\Phi}_{\mathbf{G}}$ with RPEARL PE $\mathbf{\Psi}$, for any $x \in \mathcal{M}$, w.p. $1-\delta$,

$$
\begin{aligned}
&\|\mathbf{\Phi}_{\mathbf{G}}(\mathcal{T}, \mathbf{X})(x) - \mathbf{\Phi}_{\mathcal{M}}(\mathcal{T}, f)(x)\|_2 \\
&\leq \left[C_V + 4e^{2M} C_{\mathrm{QKV}}\right] \underbrace{\left(C_{\mathcal{M}}\left(\frac{\log(1/\delta)}{N}\right)^{\frac{1}{d+4}} + \sqrt{\frac{\log(1/\delta)}{N}}\right)}_{\text{GNN bound}} \\
&\quad + \left[e^{M} C_V + 2e^{2M} C_{\mathrm{QKV}}\right] A\left(\frac{\log N}{N}\right)^{1/d}.
\end{aligned}
$$

- GNN-MNN output difference bound, adapted from Wang et al. [2024].
- $C_{\mathcal{M}}$ constant related to geometry of the manifold.
- Decays with order $\mathcal{O}\left(\left(\frac{\log(1/\delta)}{N}\right)^{1/d+4}\right)$

Sparse GT: Transferable and Computationally efficient



- Dense attention is $\mathcal{O}(N^2)$: naive scaling is impractical.
- Let k-hop neighborhood $\mathcal{N}^{\leq k}(i) = \{j : \mathrm{dist}(i,j) \leq k\}$
- Define kernel $\gamma_{ij} = \mathbb{1}_{\mathcal{N}^{\leq k}(i)} \cdot \exp\{\langle \mathbf{Q}\mathbf{x}_i, \mathbf{K}\mathbf{x}_j \rangle\}$.
- Results in **Sparse GT**:

$$\dot{\mathbf{\Phi}}_{\mathbf{G}}(\mathcal{T}, \mathbf{X})(x_i) = \mathbf{x}_i \; = \; \frac{\sum\limits_{j \in \mathcal{N}^{\leq k}(i)} \exp\{\langle \mathbf{Q}\mathbf{x}_i, \mathbf{K}\mathbf{x}_j \rangle\} \, \mathbf{V}\mathbf{x}_j}{\sum\limits_{j \in \mathcal{N}^{\leq k}(i)} \exp\{\langle \mathbf{Q}\mathbf{x}_i, \mathbf{K}\mathbf{x}_j \rangle\}}.$$

Corollary: Sparse GT + RPEARL Transferability

Corollary 4 — Transferability of Sparse GT

Sparse GT $\dot{\mathbf{\Phi}}_{\mathbf{G}}$ converges to the *restricted* MT

$$\dot{\mathbf{\Phi}}_{\mathcal{M}}(\mathcal{T}, f)(x) = \frac{\int_{\mathcal{M}} \dot{\gamma}_{xy}\, \mathbf{V} f(y)\, d\mu(y)}{\int_{\mathcal{M}} \dot{\gamma}_{xy}\, d\mu(y)}, \qquad \dot{\gamma}_{xy} = \mathbb{1}_{B_r(x)}\, e^{\langle \mathbf{Q} f(x), \mathbf{K} f(y) \rangle}.$$

If $\mathcal{N}^{\leq k}(x_i) \subseteq B_r(x_i)$, then for all x_i, w.p. $1 - \delta$,

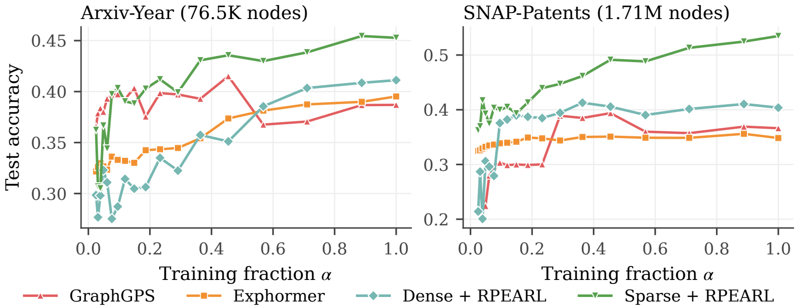
$$\|\dot{\mathbf{\Phi}}_{\mathbf{G}}(\mathcal{T}, \mathbf{X})(x_i) - \dot{\mathbf{\Phi}}_{\mathcal{M}}(\mathcal{T}, f)(x_i)\| \;\leq\; \Delta_{\mathbf{\Phi}(\mathbf{X})}.$$

- Attention mask reduces terms contributing to the integral.
- $\Delta_{\mathbf{\Phi}(\mathbf{X})}$ is the Theorem 1 GT-MT output difference.
- Sparse GT + RPEARL is size-transferable with the **same bound as Corollary 1**.

Experiments: Graph benchmarks

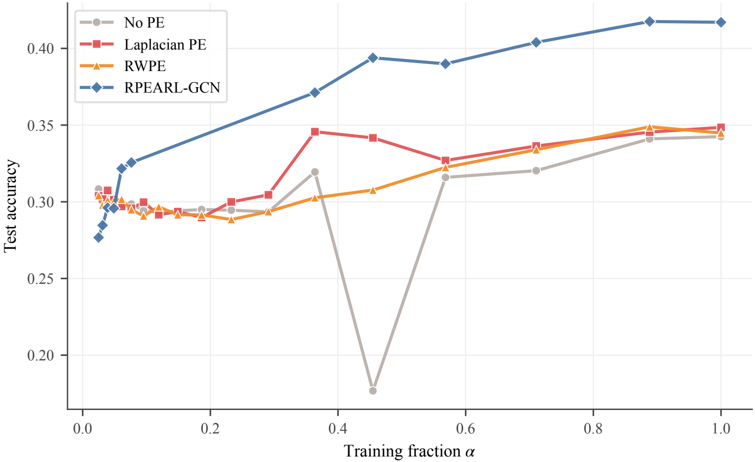
- **Datasets:** ArXiv-year, SNAP-Patents, OGBN-MAG, REDDIT-BINARY. (citation graphs, social networks)
- **Baselines:** Dense / Sparse GT + RPEARL, Exphormer, GraphGPS.
- **Setup:**
  1. Sample training graph $\mathbf{G}_{\text{Train}}$ with size $N_{\text{TR}} = \alpha \cdot N_{\text{TST}}$
  2. Train GTs on $\mathbf{G}_{\text{Train}}$.
  3. Test on $\mathbf{G}_{\text{Test}}$, with size N_{TST}
- If GT is transferable, we expect competitive performance with small values of α.

Experiments: Graph benchmarks

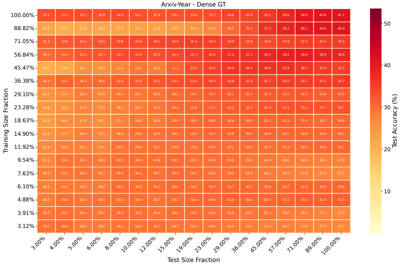


Test accuracy (↑) vs. training fraction α (proportion of test graph). Evaluation on the full-size test graph.

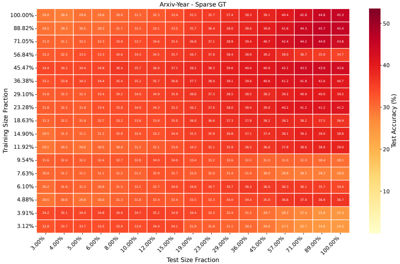
Transferability of various PEs (dense GT backbone)



Transferability of dense GT with various PEs on ArXiv-year.



Dense GT + RPEARL



Sparse GT + RPEARL

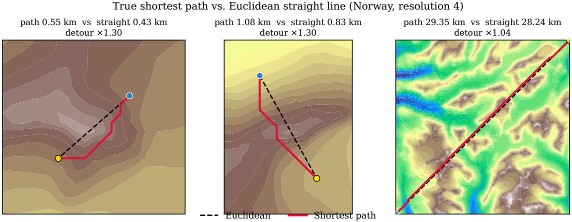
Test accuracy across (train size, test size) pairs.

Terrain manifolds application

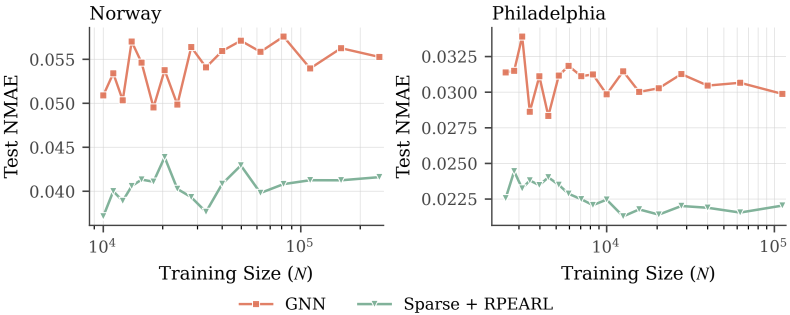
Given a terrain graph $\mathcal{M}$, and graphs $\mathbf{G}_r \sim \mathcal{M}$, where r is the stride parameter, learn an encoder for manifold points ϕ. Chen et al. [2025]

$$\mathcal{L}(s_i, d_i) = \| L_1(\phi(s_i), \phi(d_i)) - \mathrm{SPD}(s_i, d_i) \|_2 \tag{3}$$

- **Datasets:** point-cloud DEMs of **Norway** and **Philadelphia**.
- **Task:** shortest path distance (SPD) estimation via metric learning; trained end-to-end (stage one only).
- **Transferability test:** train on coarse grids with stride $r \in \{4, \ldots, 40\}$; test at full resolution ($r = 4$).



Terrain graphs: Results



NMAE (↓) vs. number of nodes at training resolution; tested at the highest resolution.

1. Consistent size generalization on both architectures with training graphs as small as 10^4 nodes.
2. Relative to GNN, on small training fractions: -25% NMAE in Norway, -26.4% NMAE in Philadelphia.

Conclusions & future work

- Graph transformers endowed with transferable positional encodings inherit their transferability.
- Various GTs exhibit competitive performance on large graphs when trained on smaller graphs.
- Confirms efficient strategies for GT training: train on small graphs, trade off global-local interaction via attention masks.
- **Future directions:**
  - Other limit models
  - Find new positional encodings that satisfy Assumption 3 (Transferable PE)

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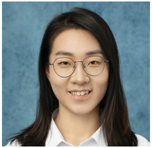

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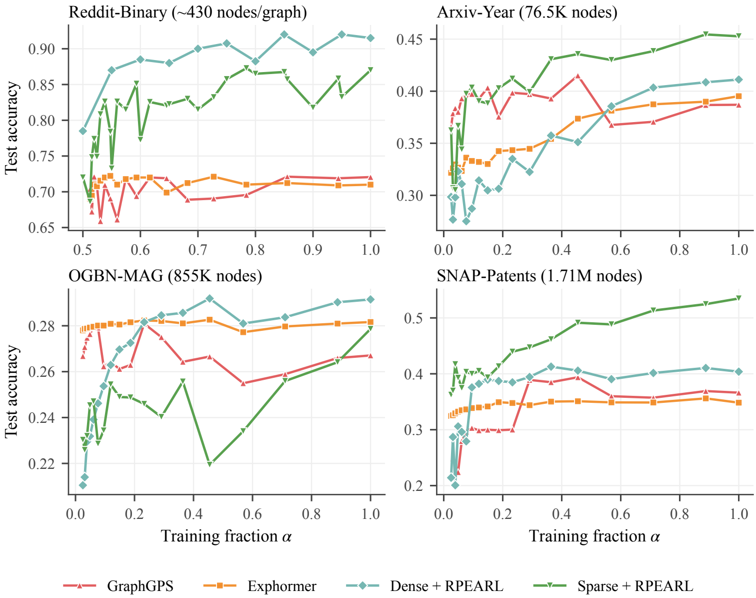
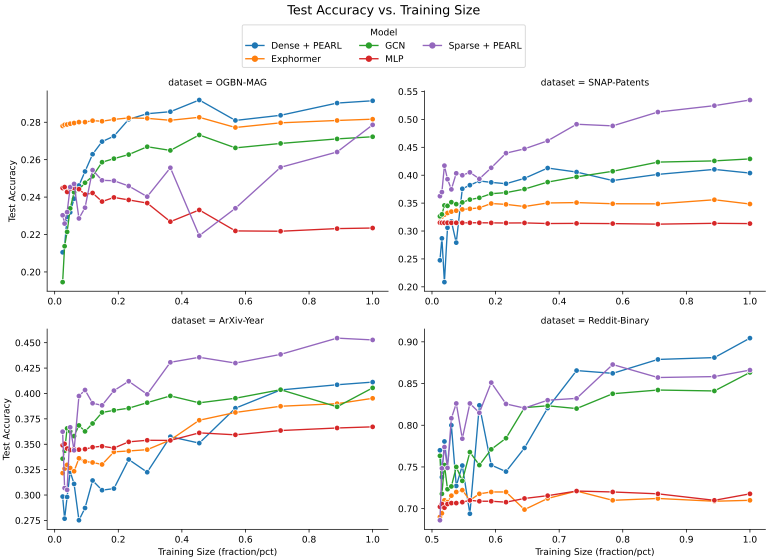
Supporting Slides

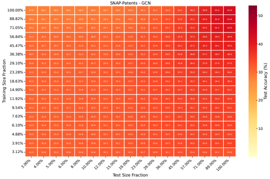
Backup: Transferability on graph benchmarks

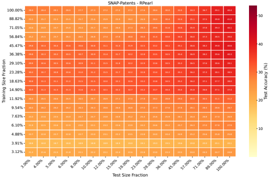


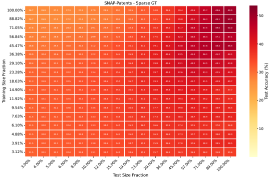
Backup: All datasets at a glance



Backup: Heatmaps: SNAP-Patents



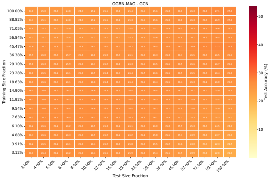




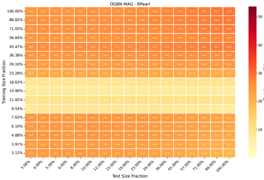
GCN | Dense GT + RPEARL | Sparse GT + RPEARL

Test accuracy across (train size, test size) pairs.

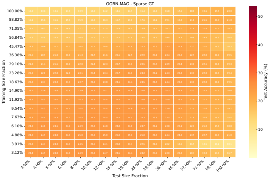
Backup: Heatmaps: OGBN-MAG



GCN



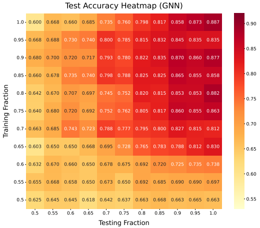
Dense GT + RPEARL



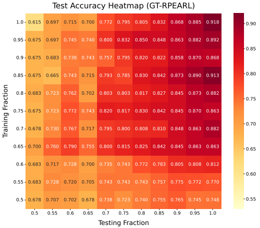
Sparse GT + RPEARL

Test accuracy across (train size, test size) pairs.

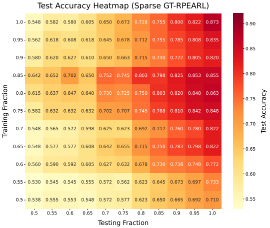
Backup: Heatmaps: REDDIT



GCN



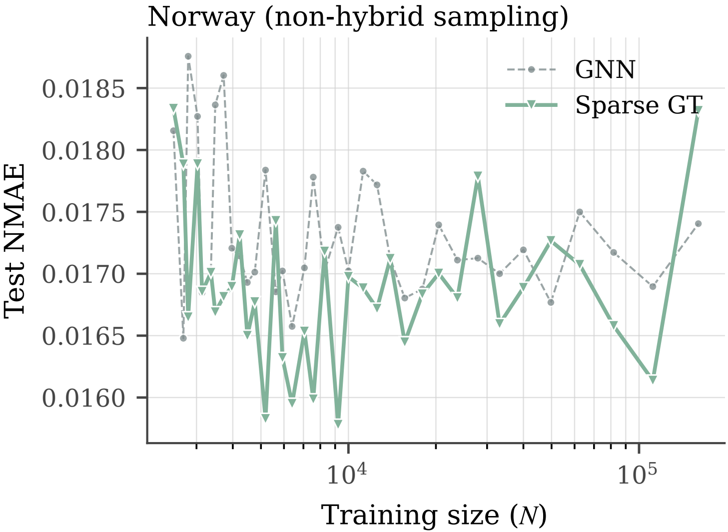
Dense GT + RPEARL



Sparse GT + RPEARL

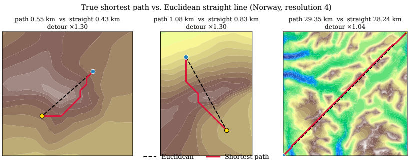
Test accuracy across (train size, test size) pairs.

First attempt: transferability on Norway

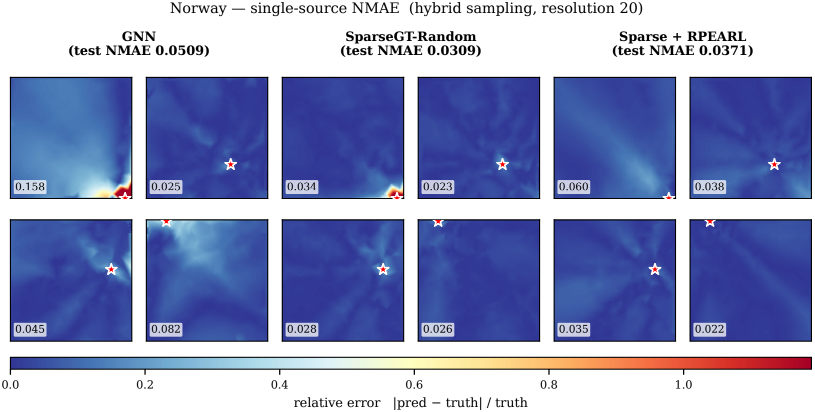


Test NMAE (↓) vs. training size; GNN vs. Sparse GT.

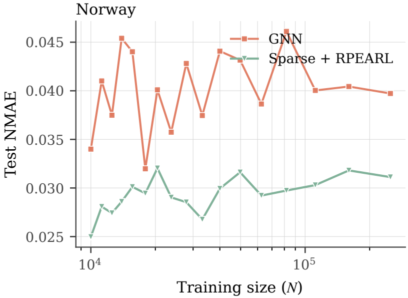
1. We observed a "representation collapse" when sampling training instances uniformly from the graph.
2. **Conjecture:** long-distance points are easily approximated by Euclidean distance, harder points not seen often enough.
3. **Solution:** bias training samples to harder points (close to mountains, nearby points.)
4. **Sampling strategy:**
   - **Sources:** 50% uniform; 50% with $p_i \propto \mathrm{Var}(\mathrm{Altitude})_i$.
   - **Targets:** 50% uniform; 50% with $p_j \propto 1/K_j$, where K_j is the hop distance from the source.



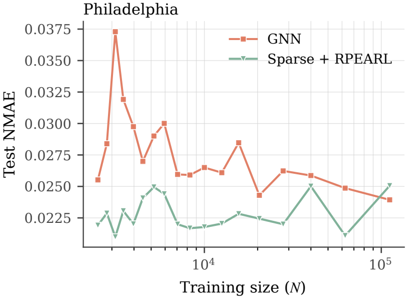
Hybrid sampling error heatmaps: Norway



Hybrid sampling: transferability (Norway & Philadelphia)



Norway



Philadelphia

Test NMAE (↓) vs. training size; GNN vs. Sparse GT + RPEARL.

References I

Mikhail Belkin and Partha Niyogi. Towards a theoretical foundation for laplacian-based manifold methods. *Journal of Computer and System Sciences*, 74(8):1289–1308, 2008.

Samantha Chen, Pankaj K. Agarwal, and Yusu Wang. De-coupled NeuroGF for Shortest Path Distance Approximations on Large Terrain Graphs. In *Forty-Second International Conference on Machine Learning*, June 2025.

Vijay Prakash Dwivedi and Xavier Bresson. A generalization of transformer networks to graphs. *arXiv preprint arXiv:2012.09699*, 2020.

Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. Graph Neural Networks with Learnable Structural and Positional Representations. 2021. URL `https://openreview.net/forum?id=wTTjnvGphYj`.

Charilaos Kanatsoulis, Evelyn Choi, Stefanie Jegelka, Jure Leskovec, and Alejandro Ribeiro. Learning Efficient Positional Encodings with Graph Neural Networks. 2024. URL `https://openreview.net/forum?id=AWg2tkbydO`.

Nicolas Keriven, Alberto Bietti, and Samuel Vaiter. Convergence and stability of graph convolutional networks on large random graphs. *Advances in Neural Information Processing Systems*, 33:21512–21523, 2020.

Sohir Maskey, Ron Levie, and Gitta Kutyniok. Transferability of graph neural networks: an extended graphon approach. *Applied and Computational Harmonic Analysis*, 63:48–83, 2023.

Luana Ruiz, Luiz Chamon, and Alejandro Ribeiro. Graphon neural networks and the transferability of graph neural networks. *Advances in Neural Information Processing Systems*, 33:1702–1712, 2020.

Luana Ruiz, Luiz FO Chamon, and Alejandro Ribeiro. Transferability properties of graph neural networks. *IEEE Transactions on Signal Processing*, 71:3474–3489, 2023.

Zhiyang Wang, Luana Ruiz, and Alejandro Ribeiro. Geometric graph filters and neural networks: Limit properties and discriminability trade-offs. *IEEE Transactions on Signal Processing*, 2024.